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MILITARY SPECIFICATION

COMPUTER, AIRCRAFT LOAD BALANCING

This specification has been approved by the Department of Defense and is mandatorg for use by the Department of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope This specification covers one type of linear slide rule load balancing computer.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on data of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-406-Plastics, Organic: General Specifications, Test Methods

MILITARY

MIL-P-116-Preservation, Methods of
MIL–P-7936-Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-T-9107-Test Reports, Preparation of
MIL-P-15047–P1astic-Whfial, Laminatad Thermosetting, Sheets, Nylon Fabric Base, Phenolic-Resin
MIL-D-70327-Drawings, Engineering and Associated Lists

STANDARDS

MILITARY

MIL-STD-130-Identification Marking of U.S. Military Property

(Copies of Specification standards drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by tbe contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning quantity production, preproduction samples shall be subjected to preproduction testing. (See 4.3.1 and 6.2.)

3.2 Components. The balancing computer shall consist of the following major components:

(a) Base.
(b) Slide.
(c) Indicator.
(d) Carrying case.
(e) Clip.

3.3 Materials. Materials which are not specifically designated herein shall be of the best quality obtainable, of the lightest practicable weight, and entirely suitable for the purpose intended.

3.3.1 Protective treatment. When materials are used in the construction of the balancing computer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protectd against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Specification MIL-D-70327.

3.5 Design.

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3.5.1 The balancing computer shall be designed to provide rapid operation for the following :

(a) Determination of the center of gravity location for any loading condition.

(b) The effect on the center of gravity of any increase, decrease, or relocation of load.

(c) The necessary shift of crew or load during fight operation to maintain the center of gravity within the permissible limits.

(d) The center of gravity limitations throughout the allowable weight range.

3.5.2 Calculations. The balancing computer design data on which the balancing computer is developed shall be based upon approved data furnished by the procuring activity or the aircraft manufacter.

3.6 construction.

3.6.1 Details of component.

3.6.1.1 Base. The base of the balancing computer shall be made of phenolic material conforming to Specification MIL-P-15047 or a vinyl copolymer resin. (Conformance with the electrical properties specified in Specification MIL-P-15047 is not required.) The base shall be recessed in such manner that the faces of the base and slide are flush with each other. The base shall be provided with grooves to carry the runners on the slide and indicator,

3.6.12 Slide. The slide shall be made of phenolic material conforming to Specification MIL-P-15047 or a vinyl copolymer resin. (Conformance with the electrical properties specified in Specification MIL-P-15047 is not required.) The slide shall be reversible and shall be supported on runners riding in grooves in the base.

3.6.1.2.1 Scales. All scales, lettering, and diagrams on the faces of the base and slide shall be printed on china white opaque vinyl copolymer resin, and shall be protected by a sheet of transparent, vinyl copolymer resin not less than 0.005 inch thick. The transparent resin shall be welded to the opaque white resin under heat and pressure. When the base and slide are made of phenolic material, the resin sheets be applied by a suitable moisture-resistant adhesive in such manner that they will not peel off, crack, or become discolored in service.

3.6.1.3 Indicator. The indicator shall be made of transparent methyl methacrylate or equal. The indicator shall be supported on runners riding in grooves in the base. The indicator shall be provided on its underneath side with an engraved, black hairline which is perpendicular to the lengthwise dimension of the balancing computer when the indicator is installed on the base. The indicator shall be so constructed that it. will maintain itself in proper alinement with the base and in such manner that the necessary friction between the indicator and the base will be provided.

3.6.1.4 Carrying case, The carrying case shall be of the scabbard type and shall be made of waterproofed natural-finish leather, reinforced waterproofed fabric, plastic, or other suitable material. When sewing is used for fabrication, seams shall be run along one side and the closed end only, and the number of stitches per inch shall be not less than 5. The open end of the carrying case shall be provided with a suitable closure as an integral part of the carrying case. When the carrying case is made of hard surfaced material which might scratch the balancing computer, it shall be lined with a soft material that will prevent scarring. The carrying case shall be designed in such manner that it will not. allow the indicator to remain lodged in the carrying case when the balancing computer is withdrawn.

3.63.4.1 A transparent window approximately 1 3/8 by 2 5/8 inches under which cards may be inserted shall be provided as an integral part of the outside of the carrying case. One card shall be furnished with each balancing computer. The card shall have spaces for entering the model designation and serial number of the aircraft and the activity to which the aircraft is assigned. The following statement shall also appear on the card: "See T.O. 1-lB-40 (AN 1-IB-4O) for Basic Weight and Index."

3.6.1.6 Chip. The clip for fastening the carring case to the aircraft shall be made of a

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suitable metal or plastic in one piece and shall be provided with three 1/8-inch holes for attachment to the aircraft.

3.7 Performance.

3.7.1 Accuracy. The accuracy of the balancing computetr shall be such that center of gravity locations determined by solving loading problems with the balancing computer shall not vary from the center of gravity locations determined_ by arithmetical calculations center of gravity more than 1/10 of 1 percent of the mean aerodynamic chord of the airplane to which the balancing computer applies.

3.7.2 Operation. The movable parts of the balancing computer shall fit together in such manner that they will slide on each other smoothly and easily without binding.

3.7.3 Environmental. The balancing computer shall be capable of operation under the following conditions (see 4.5.2) :

(a) Operating temperatures ranging from −30° to +55°C. (−22° to +131° F.).

(b) Relative humidity of 95 to 100 percent at a temperature of +550 C. (+131°F.).

3.8 Dimensions.

3.8.1 Base. The overall dimensions of the base shall not exceed 12 1/4 inches long, l 3/4 inhes wide and 13/32 inch thick.

3.8.2 Slide. The overall length of the slide shall not exceed 12 1/2 inches.

3.8.3 Indicator. The indicator shall not project more than 3/16 inch beyond the cross-sectional dimensions of the base.

3.8.4 Carrying case. The overall external dimensions of the carrying case shall not exceed 13 inches long, 2 3/4 inches wide and 1 1/2 inches thick.

3.8.5 Weight. The total weight of the balancing computer, including carrying case and clip, shall not exceed 1 pound,

3.9 Arrangement,

3.9.1 Scales. The location and arrangement of scales on the balancing computer shall be such that a balance calculation may be executed with a minimum number of movements. All scales shall be appropriately labeled with the compartment, fuel, or oil tark or useful load item to which they apply. Scales shall not be bunched in a manner which will produce a crowded appearance, and the direction in which the indicator is to be moved over the scale when adding weight shall be indicated by an arrow at the zero end of each scale or tower of scales. The balancing computer shall be arranged so that movement of the indicator from right to left will represent a forward movement of the

3.9.2 Center of gravity limits. The center of gravity limits on the base of the balancing computer shall indicate the variation accompanying changes in gross weight. For the initial design, the center of moments for the balancing computer shall be taken at the bit nearest which the center of gravity will fall under normal loadings in order that this limit may be represented by a vertical line. A chart of gross weight versus center of gravity position in percent mean aerodynamic chord (MAC) and inches from the reference datum shall be shown in the recessed portion of the base. The following color coding shall be used:

(a) Red areas shall indicate regions in which readjustment of the load is required.

(b) Yellow areas at the forward or aft end of the center of gravity range shall indicate regions in which restrictions resulting from aerodynamic or hydrodynamic conditions are applicable. The corresponding area on the grid in the recessed portion of the base shall be colored yellow and appropriate notations made.

(c) A white area shall indicate the region in which safe balance is obtained. The region shall be labeled “Loading Range” and shall contain the following note in red letters: “Check Landing Condition: Use of Expendable Load Causes CG Change?’ Applicable statements shall be used to identify area limits. The following are some of those considered acceptable:

FWD Limit-Gear Down
FWD Limit (Percent UC)

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FWD Limit-Gear Down (1 Percent MAC)
FWD Hydro. Limit Add (1 Percent MAC) if constant
Aft Limit (1 Percent MAC)
Aft Limit-Gear Down
Aft Hydro. Limit. Add (1 Percent) if constant

The following note shall be shown in a yellow box on or adjacent to the gross weight versus center of gravity chart: "See Applicable Flight Handbook or Technical order for Gross Weight Restrictions." The points of demarkation between the white and red areas, white and yellow areas, and yellow and red areas shall be indicated by the use of a single black line of the same weight as those used to show the various center of gravity positions in percent MAC.

3.9.3 Index formula and scales. The formula by which the index setting may be calculated using the basic weight and basic moment shall be shown on the balancing computer. Unless otherwise specfied, the balancing computer shall also be provided with scales by which the index setting may be determined using the basic weight and basic moment. These scales shall be placed on the back of the slide. The index scale on the base of the computer shall extend over the entire center of gravity range, including restricted areas, and shall cover as much of the length of the balancing computer as is practicable. The numbers on the index scale shall increase from left to right, and negative numbers shall not be included on the scale. The following note in red letters shall appear just above the right end of the index scale:

"CAUTION: Check basic index to be sure index formula for this balancing computer has been used."

3.9.4 Interior arrangement diagram. A diagram showing the interior arrangement of the aircraft to which the balancing computer applies shall be shown on the back of the base. The diagram shall be in sufficient detail to permit ready identification of the various stations and compartments, and shall show a horizontal distance scale.

3.9.5 Gross weight scale. A gross weight scale to facilitate reading the chart of gross weight versus center of gravity position shall be placed at the left end of the slide. This gross weight scale shall be so located that it will coincide with the gross weight scale cm the recessed portion of the base when viewed in a line perpendicular to the face of the balancing computer.

3.9.6 Alinement.Suitable previsions shall be incorporated in the balancing computer whereby the alinement of the several resin sheets applied to the base maybe easily verified and the scales checked for distortion.

3.10 Identification of product Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.10.1 Computer. The following shall be shown on the base of the balancing computer:

(a) Aircraft model designation.
(b) US.
(c) The manufacturer's serial number of the balancing computer. This number shall be stamped on one end of the base and the slide. Numbers shall be assigned by the contractor and approved by the procuring activity.

3.10.2 Carring case. The following shall be shown on the carrying case: US.

3.11 Engineering sample plate. A suitable full-size reproduction of the faces of the base and slide illustrating the arrangement of the balancing computer which the contractor intends to furnish under the contract shall be submitted to the aircraft manufacturer and procuring activity for approval before the balancing computer is put into production. A preliminary full-size faceplate shall be submitted for each aircraft model for which balancing computers are to be furnished. When a revised balancing computer is to be furnished, a preliminary balancing computer faceplate illustrating the proposed revision shall be submitted.

3.12 Instruction book One instruction book shall be furnished with each balancing computer. This book shall contain detailed in-

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structions for using the balancing computer, the solutions of illustrative sample problems, and instructions for determining the index from the basic weight and moment. Appropriate illustrations shall be included in the book to improve its clarity and effectiveness. A sample instruction book for each balancing computer design shall be submitted to the aircraft manufacturer and the procuring activity for approval prior to printing.

3.13 Workmanship. Workmanship shall be neat and accurate and shall be in accordance with high-grade commercial manufacturing practice for slide rules.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government inspector as specified in the contract or order.

4.2 Classification of tests. The inspection and testing of the balancing computers shall be as follows:

(a) reproduction tests (see 4.3).
(b) Acceptance tests (see 4.4).

4.3 Preproduction tests.

4.3.1 Production test samples. The preproduction test samples shall consist of a model representative of the production equipment. It shall be tested at a site designated by the procuring activity or, when so stated in the contract, at the contractor's plant under supervision of the procuring activity.

4.3.2 Preproduction test reports. When the contractor performs the preproduction tests, he shall prepare a preproduction test report in accordance with Specification MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.3.3 Tests. Preproduction tests shall consist of all the tests described under 4.5.

4.3.4 When specified in the contract, preproduction tests may be waived if the contractor's item has previously passed a preproduction test specified herein.

4.4 Acceptance tests. Accaptance tests shall consist of individual tests and periodic sampling plan and tests.

4.4.1 Individual tests. Each balancing computer shall be visually examined for workmanship and mechanical operation.

4.4.2 Periodic sampling plan and test. One balancing computer shall be selected at random by the inspector from the first 10 units manufactured on a production contract and subjected to the tests described in 4.5.1, 4.5.2, and 4.5.3. Thereafter one balancing computer shall be selected at random from each additional 100 computers manufactured on that contract, and subjected to the tests described in 4.5.1, 4.5.2, and 4.5.3.

4.5 Test methods.

4.5.1 Examination of product. The test computer, including each major component, shall be examined to determine compliance with the applicable requirements specified herein with respect to materials, finish, workmanship, marking, dimensions, arrangement, operation, and weight. The weight shall be determined on any suitable scale which has been accurately calibrated.

4.5.2 Functional tests. The test computer shall be test operated to determine compliance with the requirements of 3.7.1 Typical loading problems shall be computed to allow determination of computer accuracy against calculated mathematical values. The center of gravity location represented by the position of the indicator at the completion of the computer calculation shall be determined by substituting the gross weight and final index setting in the index formula, solving for the moment, and determining the arm. The arm thus obtained shall then be compared with the arm obtained by arithmetical calculation. The moment arms used in the arithmetical calculation shall coincide with the moment arms used for the scales of the balancing computer.

4.5.3 Envirmental test. The computer shall be tested in accordance with "Accelerated

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Service Tests (Temperature and Humidity Extremes)" method 6011 of Specification L-P-406 to determine compliance with the requirements of 3.7.3.

4.5.4 Accelerated service tests. The test computer inserted in the open case shall be tested in accordance with method 6011, "Accelerated Semite Tests (Temperature and Humidity Extremes)" the "High-low temperature test" of Specification L-Pl406. The tests shall consist of 3 cycles and shall be terminated in the cold condition. The test temperatures shall be + 175° F. and -70°F. (+80°C. and -57° c.).

4.5.5 Packing und marking. The packing and marking of the computers shall be inspected to determine conformance to this specification.

4.6 Rejection and retest. When one item seleced from a production run f ails to meet the specification no items still on hand or later produced shall be accepted until the extent and cause of failure are determined,

4..6.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Each balancing computer, including carrying case, instruction book, and clip shall be prepared for delivery in accordance with Specification MIL-P-7936. Unless otherwise specified, preservation and packaging shall be in accordance with level A and packing in accordance with level B. Level A preservative and packaging shall be method III of Specification MIL-P-116.

6. NOTES

6.1 Intended use. The balancing computer covered by this specification is intended for use in determining longitudinal center of gravity positions of aircraft.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Levels of preservation and packaging, and packing. (See section 5.)

(c) Applicable aircraft designation and model.

(d) Designation of test site for preproduction testing.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifcations, or other data is not to be regarded by implication or otherwise as in any marine? licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be reiated thereto.

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